



**Federal Aviation
Administration**

Initial En Route Qualification Training

**Lesson 18
Lateral Separation**

Course 50148001

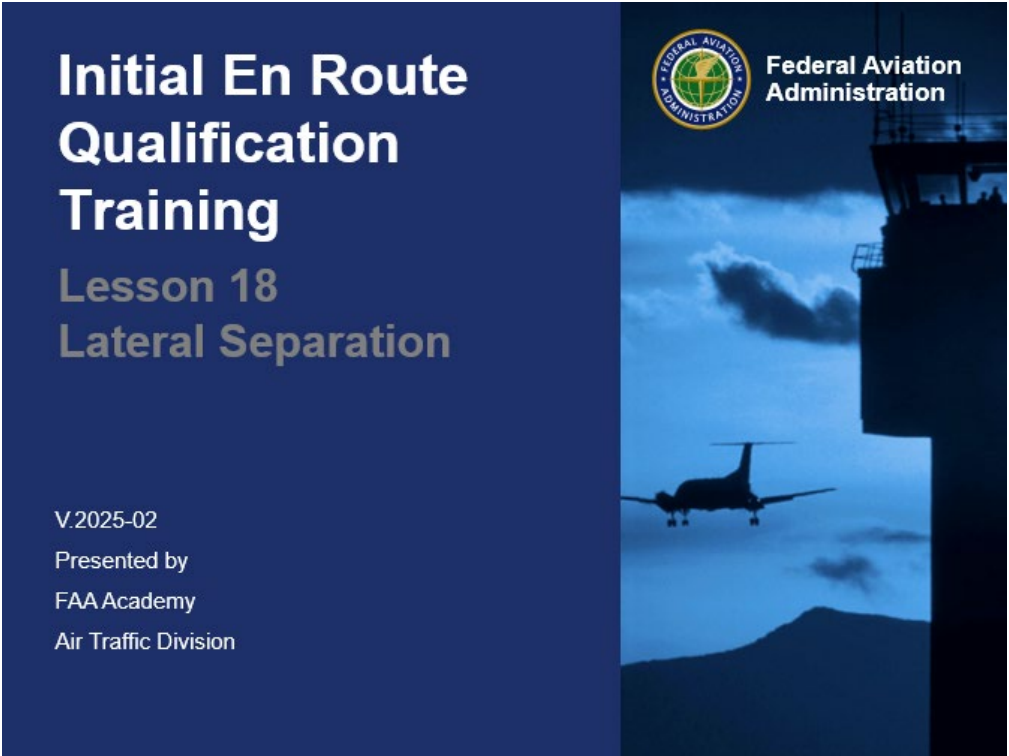
LESSON PLAN DATA SHEET

COURSE NAME:	INITIAL EN ROUTE QUALIFICATION TRAINING
COURSE NUMBER:	50148001
LESSON TITLE:	LATERAL SEPARATION
DURATION:	12+00 HOURS
DATE REVISED:	2025-02
VERSION:	V.2025-02
REFERENCE(S):	FAA ORDER JO 7110.65, AIR TRAFFIC CONTROL; AERO CENTER MAP; FAA ORDER 8260.3 UNITED STATES STANDARD FOR TERMINAL INSTRUMENT PROCEDURES (TERPS)
HANDOUT(S):	lat1.f2k, lat2.f2k, lat3.f2k, lat4.f2k, lat5.f2k, latptask.f2k - LATERAL SEPARATION PART-TASK STRIPS
EXERCISE(S)/ ACTIVITY(S):	EXERCISE 1: APPLYING LATERAL SEPARATION EXERCISE 2: DEMONSTRATING LATERAL SEPARATION BEST PRACTICES ACTIVITY: ENSURING SEPARATION
END-OF-LESSON TEST:	YES
PERFORMANCE TEST:	NONE
MATERIALS:	NONE
OTHER PERTINENT INFORMATION:	APPENDIX A: ZAE HOLDING PATTERNS

DISCLAIMER

Air Traffic training materials are provided by the Federal Aviation Administration Office of Technical Training. The training materials are subject to revision, amendment and/or corrections. The materials are to be used for the intended purpose only and are not intended for distribution beyond the purpose of training. Updates (if required) shall be provided through a formal distribution process. Unsolicited updates will not be processed.

INTRODUCTION



The image shows the cover of a presentation titled "Initial En Route Qualification Training Lesson 18 Lateral Separation". The cover has a dark blue background. On the left, the title is written in white and yellow text. On the right, there is a photograph of an airplane flying in a blue sky with clouds, and a portion of an air traffic control tower is visible on the far right. The FAA logo and name are in the top right corner. Below the title, the version number "V.2025-02" and the presenter "Presented by FAA Academy Air Traffic Division" are listed.

**Initial En Route
Qualification
Training**

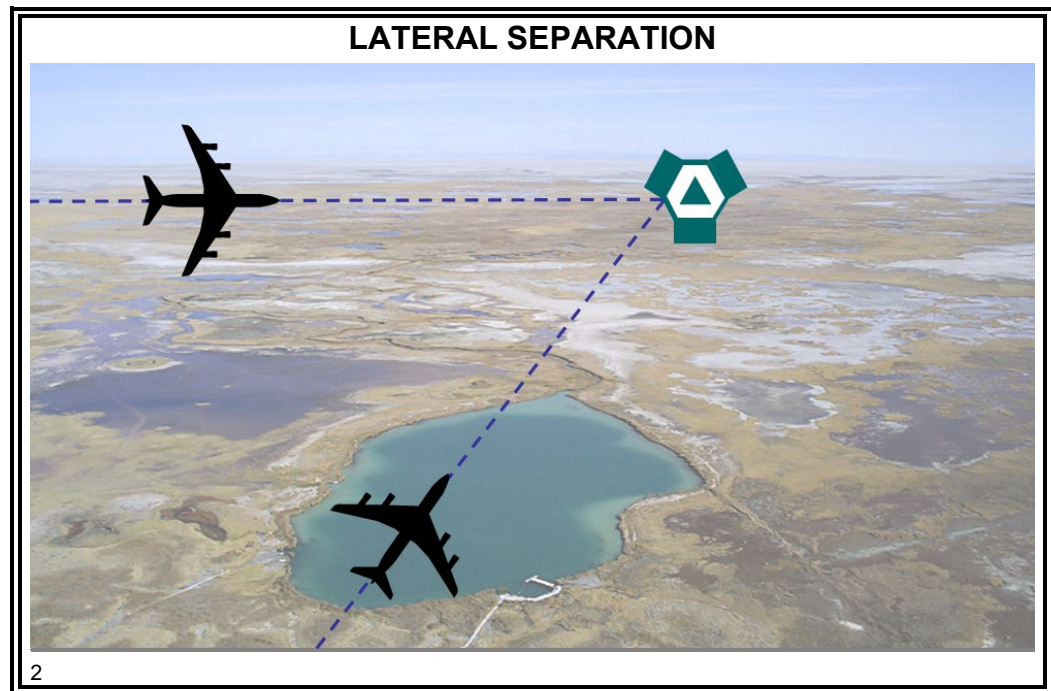
**Lesson 18
Lateral Separation**

V.2025-02
Presented by
FAA Academy
Air Traffic Division

1

In the previous lesson, you were introduced to longitudinal separation. However, you will **not** be able to apply vertical separation in all situations. Air Traffic Controllers generally use a combination of separation rules to achieve a safe operation. A combination of lateral and vertical separation will allow you to separate aircraft more efficiently.

INTRODUCTION *(Continued)*



Your ability to apply lateral separation in conjunction with vertical separation will help resolve traffic situations as they become more complex throughout your training.

Purpose

This lesson will cover lateral separation and its application in air traffic control situations.

Lesson Objectives

LESSON OBJECTIVES

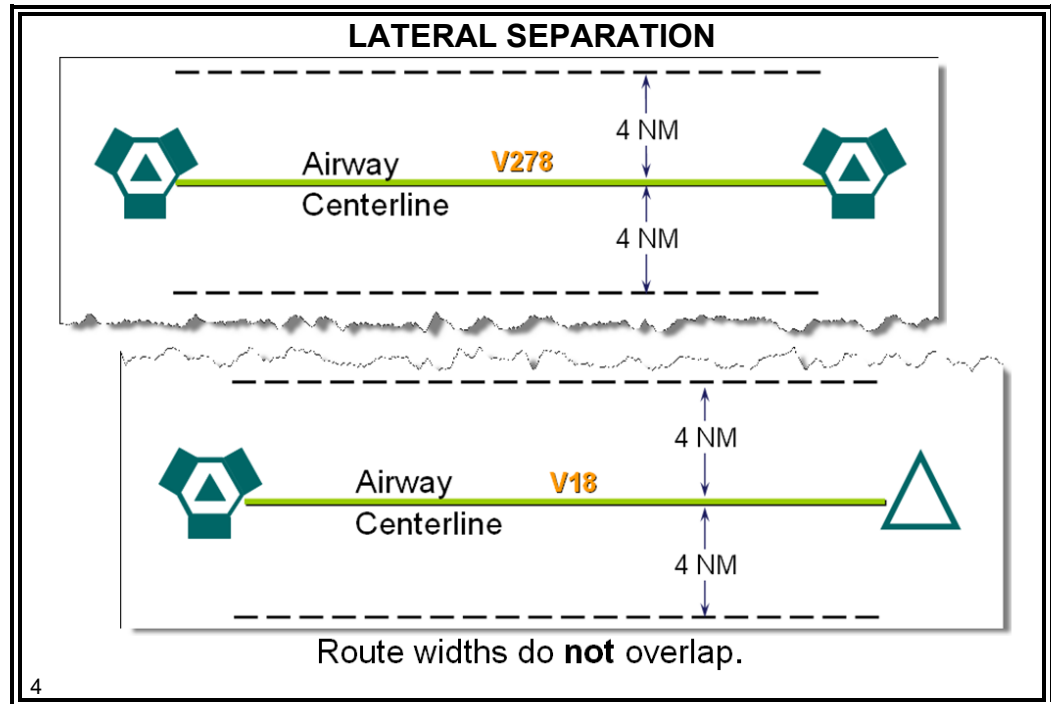
- On an End-of-Lesson Test, and in accordance with FAA Order JO 7110.65, you will identify:
 - Airspace to be protected along airways/routes, and holding patterns
 - Minima and procedures for applying lateral separation

3

LATERAL SEPARATION METHODS

Methods

JO 7110.65,
par. 6-5-1



- ☉ Clear aircraft on different airways or routes whose widths or protected airspace **must not** overlap.

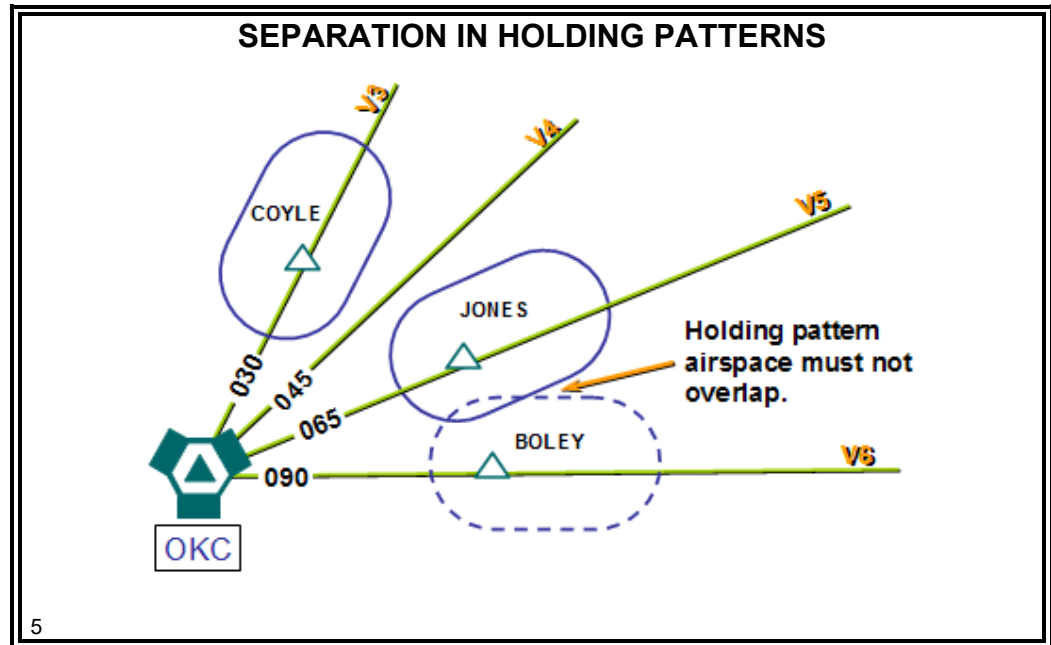
NOTE: Protected airspace can touch, but it **cannot** overlap.

Continued on next page

LATERAL SEPARATION METHODS *(Continued)*

Methods (Cont'd)

JO 7110.65,
par. 6-5-1



Clear aircraft to hold over different fixes whose holding patterns **must not** overlap.

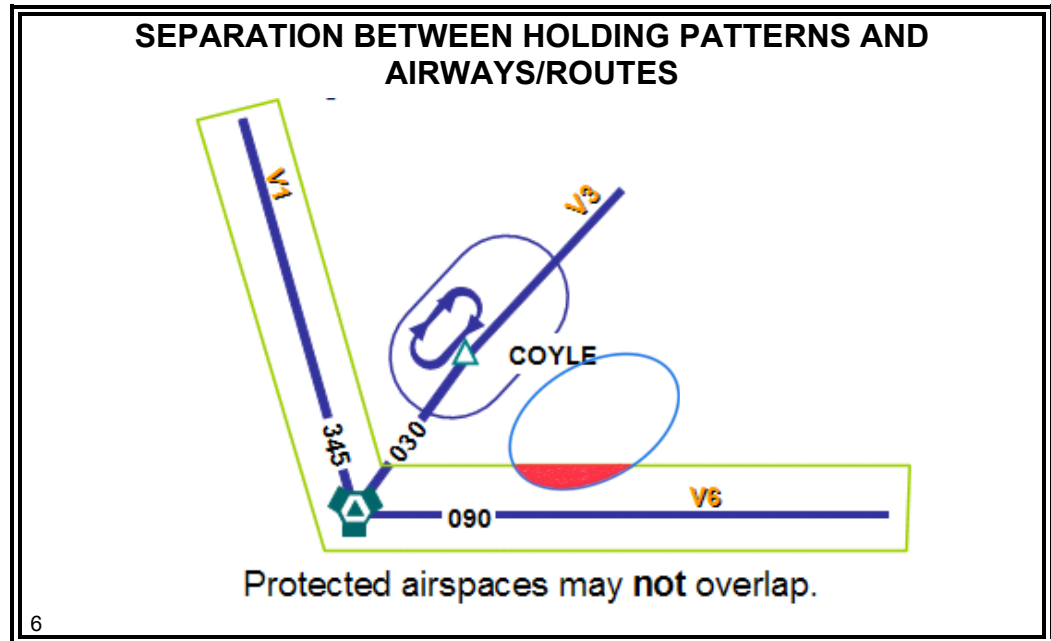
- If holding patterns overlap, do **not** hold aircraft at the same altitude

Continued on next page

LATERAL SEPARATION METHODS *(Continued)*

Methods (Cont'd)

JO 7110.65,
par. 6-5-1



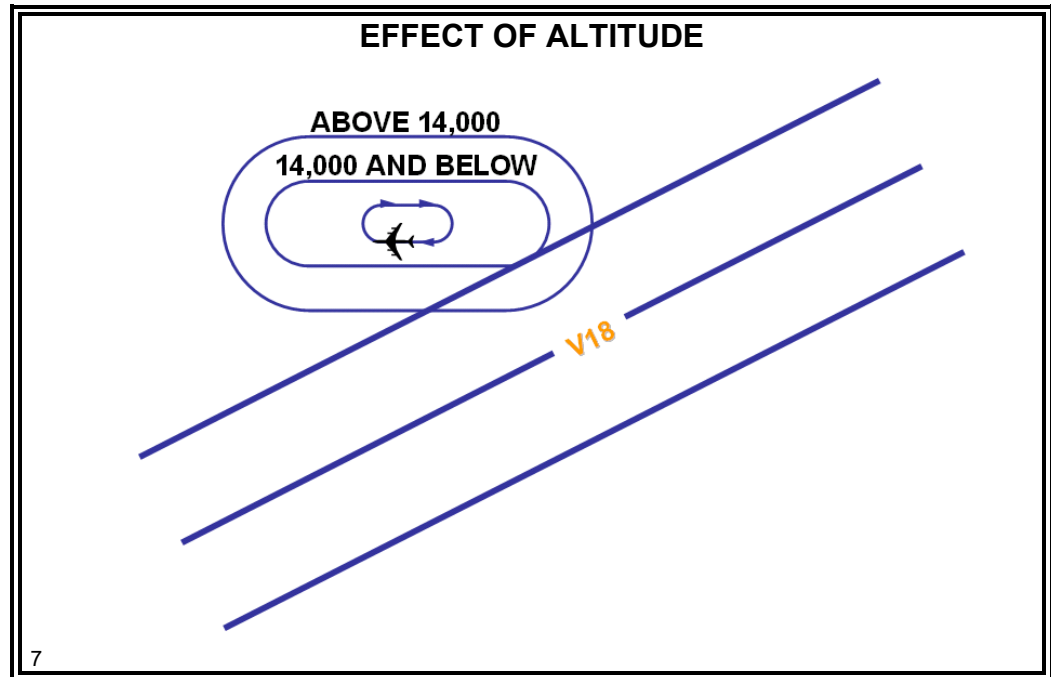
- ☉ Clear aircraft to hold over different fixes whose holding patterns **must not** overlap other protected airspace.

Continued on next page

LATERAL SEPARATION METHODS *(Continued)*

Methods (Cont'd)

JO 7110.65,
par. 6-5-1



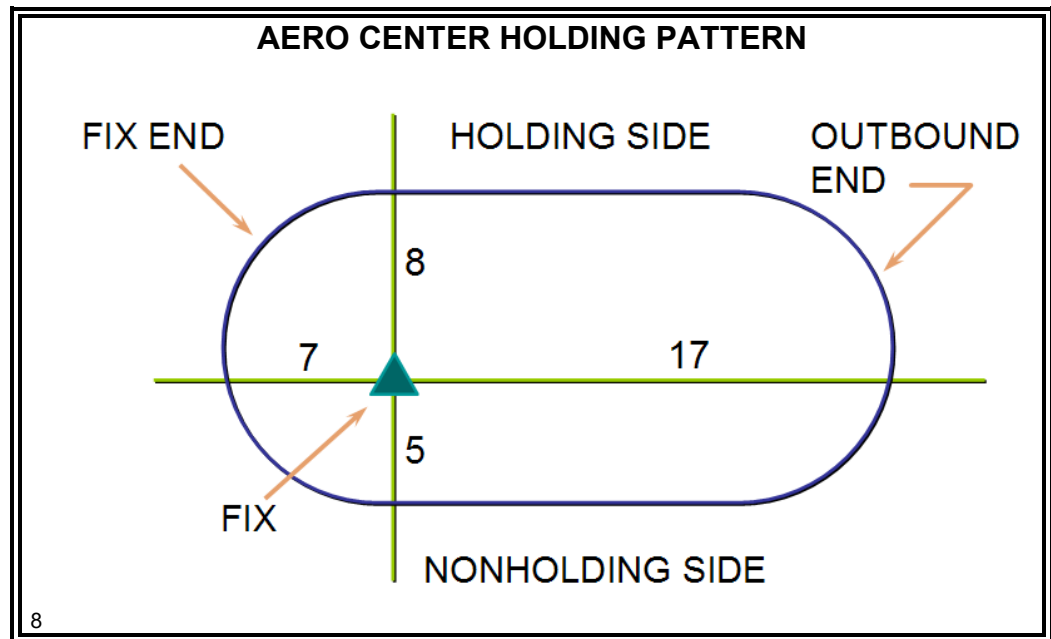
- Holding pattern protected airspace area increases with an increase in an aircraft's altitude and/or speed.

Continued on next page

LATERAL SEPARATION METHODS *(Continued)*

Methods (Cont'd)

JO 7110.65,
par. 6-5-1



- The above holding pattern was used in Aero Center to determine the protected airspace used in nonradar maps, scenarios, and procedures

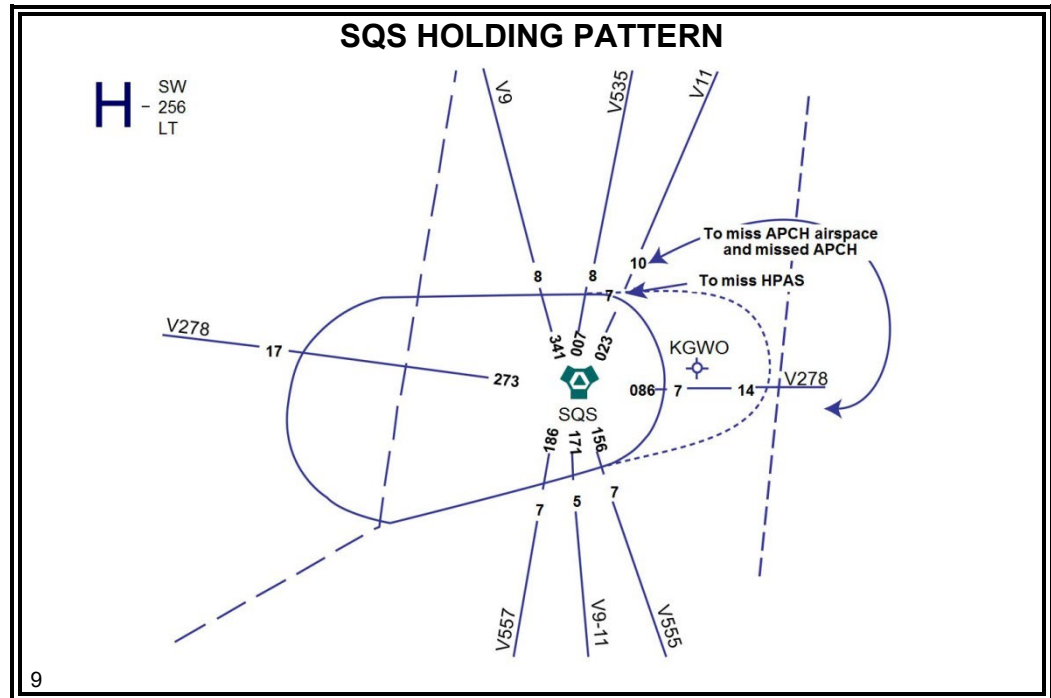
NOTE: Other templates are available for other altitudes and types of aircraft.

Continued on next page

LATERAL SEPARATION METHODS *(Continued)*

Methods (Cont'd)

JO 7110.65,
par. 6-5-1



NOTE: SQS holding pattern overlaps Sector 67 airspace and requires coordination before and after use. The dotted line at the airport end depicts additional protected airspace for the actual approach to the airport including the missed approach. Other Aero Center holding patterns are in Appendix A of this lesson.

NOTE: KGW0 departures which require a crossing restriction or report on V11, V278 or V535 of 10 miles NE SQS or less must include "ON/established on" an airway.

Example: "Cross one zero miles NE SQS VORTAC ON/Established ON V11 at or below 6000"....

LATERAL SEPARATION METHODS *(Continued)*

Knowledge Check

KNOWLEDGE CHECK

❖ **QUESTION:** The protected airspace of a holding pattern must **not** _____ of an airway.

- Ⓐ overlap the protected airspace
- B. overlap the centerline
- C. touch the protected airspace

10

KNOWLEDGE CHECK

❖ **QUESTION:** The size of the protected airspace of a holding pattern varies according to _____ and _____.

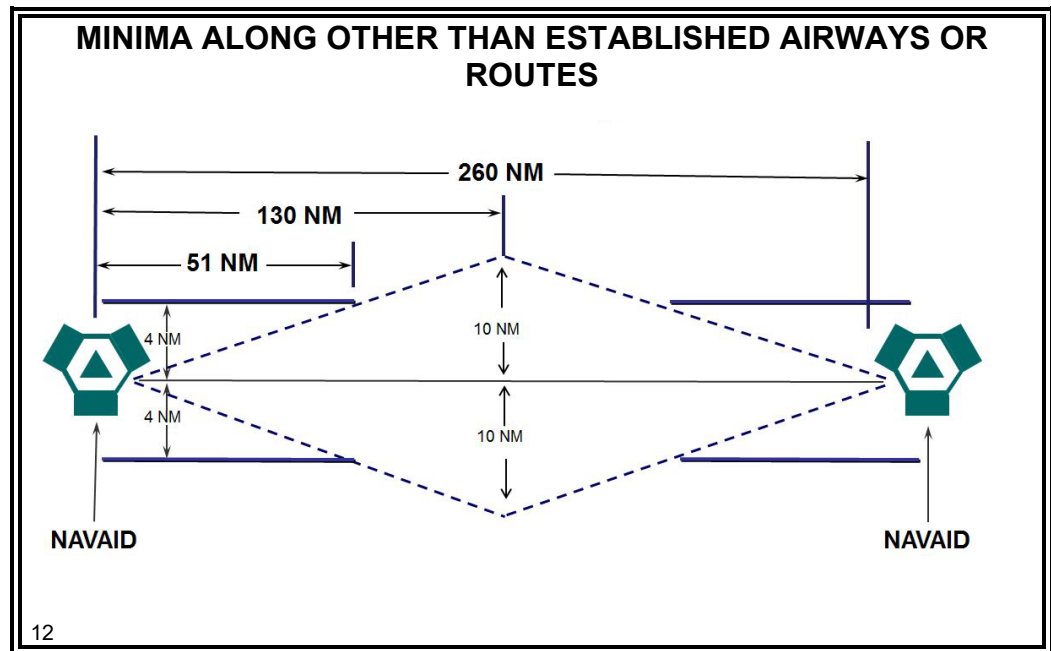
- A. speed; direction of holding
- B. altitude; wind speed
- Ⓒ speed; altitude

11

PROTECTED AIRSPACE

Minima Along Other Than Established Airways/Routes

JO 7110.65,
par. 6-5-4



Protect airspace along other than established airways or routes as follows:

- ⊙ Direct courses and course changes of 15 degrees or less:
 - Via NAVAIDs or radials FL 600 and below- 4 miles on each side of the route to a point 51 miles from the NAVAID, then increasing in width on a 4 1/2 degree angle to a width of 10 miles on each side of the route at a distance of 130 miles from the NAVAID.

PROTECTED AIRSPACE *(Continued)*

Protected
Airspace

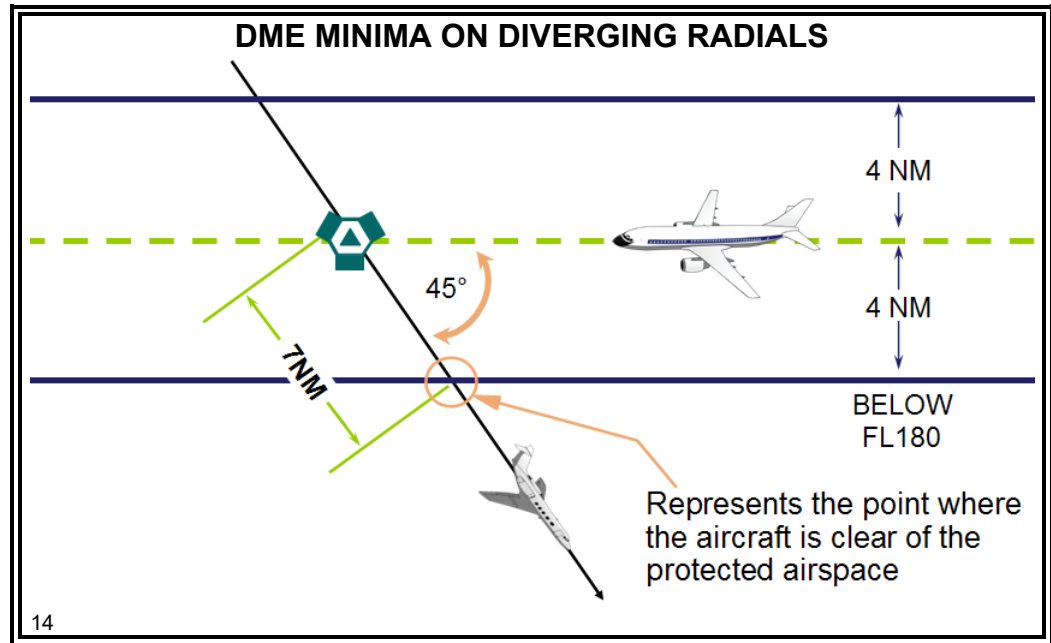


NOTE: This animation is intended to illustrate the concept of lateral and longitudinal protected airspace.

MINIMA ON DIVERGING RADIALS

Separation Criteria

JO 7110.65,
par. 6-5-2,
table 6-5-2



- ⊙ Consider separation to exist between aircraft:
 - Established on diverging radials of the same NAVAID
 - At least 15 degrees divergence when either aircraft is clear of the airspace to be protected for the other aircraft

MINIMA ON DIVERGING RADIALS *(Continued)*

DME Divergence Distance Minima

JO 7110.65,
par. 6-5-2,
table 6-5-2

DME DIVERGENCE DISTANCE MINIMA

Divergence (Degrees)	Distance (NM) (below FL180)
15	17
20	13
25	11
30	9
35	8
45	7
55	6
90	5

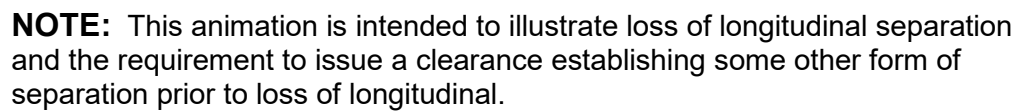
15

☉ DME Application Table

- Used to determine the distance required for angles of divergence to clear protected airspace
 - If the divergence falls between two values, use the greater distance minima
- Compensates for DME slant-range error
 - Slant-range is the line-of-sight distance between the aircraft and NAVAID

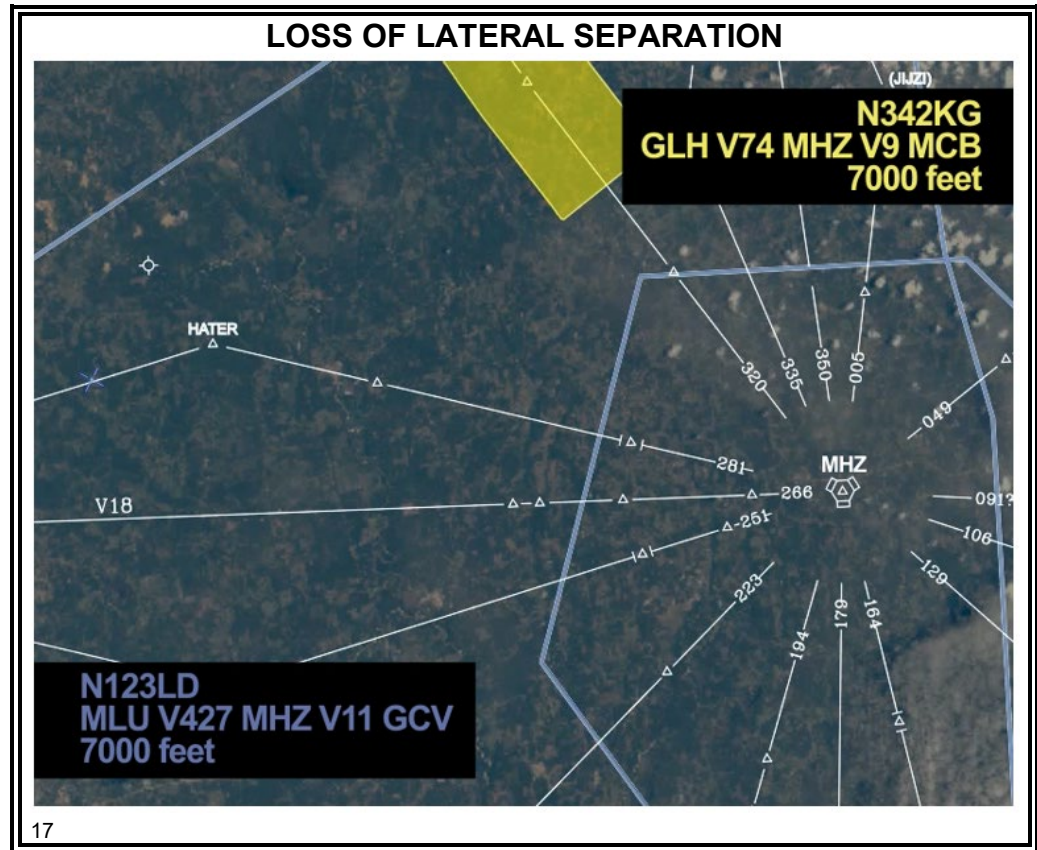
NOTE: Students must memorize DME Divergence Table. It will **not** be available during tests.

Loss of Longitudinal Separation



MINIMA ON DIVERGING RADIALS (Continued)

Loss of
Lateral
Separation



NOTE: This animation is intended to illustrate loss of lateral separation.

MINIMA ON DIVERGING RADIALS *(Continued)*

Establishing
Vertical
Separation

ESTABLISHING VERTICAL SEPARATION WITH A CROSSING RESTRICTION



"N123LD cross 8 miles northwest Magnolia VORTAC at or above eight thousand. Climb and maintain nine thousand."

18

NOTE: This animation is intended to illustrate the establishment of vertical separation at the same point where lateral separation is lost.

MINIMA ON DIVERGING RADIALS *(Continued)*

Knowledge Check

KNOWLEDGE CHECK

❖ **QUESTION:** To determine a distance for lateral separation when the degrees divergence falls between two values listed in the divergence table, _____.

- A. take the average of the two distances
- Ⓐ use the greater distance
- C. use the lesser distance

19

KNOWLEDGE CHECK

❖ **QUESTION:** The width of V18, 40 miles east of MLU VORTAC, is _____ miles.

- A. 4
- Ⓑ 8
- C. 10

20

Continued on next page

MINIMA ON DIVERGING RADIALS *(Continued)*

Knowledge Check (Cont'd)

KNOWLEDGE CHECK

45°

4NM

4NM

? NM

Represents the point where UAL1468 must be at 17,000 feet

UAL1468
H/B744/A
↑170

N25MH
160

QUESTION: How many miles southeast of the VORTAC must UAL1468 (DME aircraft) be at 17,000 feet to provide minimum lateral separation from N25MH? >

21

EXERCISE 1: APPLYING LATERAL SEPARATION

Exercise 1



APPLYING LATERAL SEPARATION EXERCISE



Purpose: to practice applying lateral separation rules

Directions: use lateral separation rules and provided aides to answer questions

22

Directions

Use your Aero Center Map to answer questions 1 through 10 in this exercise.

Questions

❖ **QUESTION 1:** What is the required mileage for the following diverging angles? (Use the DME Application Table on p. 13)

DIVERGING ANGLES	BELOW FL180
15 degrees	17
45 degrees	7

❖ **QUESTION 2:** AAL42 is 40 miles west of Magnolia VORTAC on V427 at one six thousand. At this position, what airspace **must** be protected on either side of the route?

4 mi

23

Continued on next page

EXERCISE 1: APPLYING LATERAL SEPARATION

(Continued)

Questions (Cont'd)

❖ **QUESTION 3:** Under what condition may aircraft be cleared to hold over different fixes at the same altitude?

If the holding patterns do NOT overlap

❖ **QUESTION 4:** When are aircraft established on radials of the same NAVAID that diverge by at least 15 degrees considered to be laterally separated?

*Beyond the published
encompass of the other
course*

24

Use the following flight progress strip on DAL660 to answer questions 5 through 10.

Question 5 Progress Strips

DAL660 H/B763/I T490 66 03	GLH 1255	59 12 SQS	130	IGB →	KTXK V278 IGB LGC LGC1 KATL	
N401L LJ25/A T460 66 01	EDC XXXX	T→SW- SQS XXXX KGWO P1250	↑ 150 X 6 SE SQS + 120 ↑ ASSUMING COM'S MODE IS INACTIVE	MHZ ↘	KGWO SQS V555 KMCB/0025	D-A

25

❖ **QUESTION 5:** TIME: 1250 RUNWAY: 23

To provide separation from DAL660, N401L **must** have a crossing restriction of 6 miles southeast of SQS VORTAC at or below 120 feet.

Continued on next page

EXERCISE 1: APPLYING LATERAL SEPARATION

(Continued)

Questions (Cont'd)

Question 6 Progress Strips						
DAL660 H/B763/I T490 66 03	GLH 1255	59 12	130	IGB	KTXK V278 IGB LGC LGC1 KATL	
		SQS				
N201PB LJ24/A T440 66 03	UJM 1239	50 12	170✓ 170/24NW X 6 NW ↓ 11g x SQS @ 70	KGWO 1257	KMEM UJM V9 SQS KGWO/1255 C _{SQS}	V/R 1248 H- 100 100 100
		50				
		SQS				

26

QUESTION 6: TIME: 1248

To provide separation from below DAL660, N201PB **must** have a crossing restriction of 6 miles northwest of SQS VORTAC at or below 70 feet.

Continued on next page

EXERCISE 1: APPLYING LATERAL SEPARATION

(Continued)

Questions (Cont'd)

Question 7 Progress Strips						
DAL660 H/B763/I T490 66 03	GLH 1255	59 12 SQS	130	IGB	KTXK V278 IGB LGC LGC1 KATL	
N401L LJ25/A T460 66 01		T→SW SQS KGWO P1250	↑ 150 X 6 56 SQS ↓ 120	MHZ	KGWO SQS V555 KMCB/0025	D-A

27

❓ **QUESTION 7:** To provide lateral separation from DAL660, N401L **must** have a crossing restriction of 6 miles southeast of SQS VORTAC at or below 70 feet.

Question 8 Progress Strips						
DAL660 H/B763/I T490 66 03	GLH 1255	59 12 SQS	130	IGB	KTXK V278 IGB LGC LGC1 KATL	
N751L LJ55/A T465 66 02	MHZ 1249	57 12 SQS	130✓	HLI	KMCB V555 SQS V11 HLI M41/1312	

28

❓ **QUESTION 8:** When will separation cease to exist between N751L and DAL660?
1249

Continued on next page

EXERCISE 1: APPLYING LATERAL SEPARATION

(Continued)

Questions (Cont'd)

Question 9 Progress Strips						
DAL660 H/B763/I T490 66 03	GLH 1255	59 12	130	IGB	KTXX V278 IGB LGC LGC1 KATL	
		SQS				
N50JC GLF3/A T450 66 02	MHZ 1249	57 12	$\uparrow 160$ $\times 6 SE SQS$ $\uparrow 140'$	UJM	KJAN MHZ V9 UJM KMEM/1312	
		SQS				

29

❖ **QUESTION 9:** TIME: 1255 RUNWAY: 18

To provide lateral separation from DAL660, N50JC **must** have a crossing restriction of 6 miles southeast of SQS VORTAC at or above 140 feet.

Question 10 Progress Strips						
A25817 BE55/B T230 66 02	STUEE 0040	57 12	$\downarrow$ $90 \downarrow 60$	KJAN	KMLU V18 MHZ KJAN	H ^{NW}
		58				
		MHZ				
N256B BE65/A T160 66 01			$\uparrow 90$ $\times 9 SW MHZ$ $\downarrow 50'$	KHEZ	KJAN MHZ V245 KHEZ/0030	
		KJAN P0100				

30

❖ **QUESTION 10:** TIME: 0050

To provide separation from A25817's holding pattern at MHZ, N256B **must** be restricted to cross 9 miles southwest of MHZ VORTAC at or below 50 feet.

LATERAL SEPARATION FROM SPECIAL USE AIRSPACE

Conflict with
MOA

CONFLICT WITH MOA



"N342KG Cleared to Texarkana airport via direct Sidon Victor 278. Climb and maintain one zero thousand. B-G."

31

NOTE: This animation is intended to illustrate loss of separation between GWO departure and the CBM MOA.

LATERAL SEPARATION FROM SPECIAL USE AIRSPACE *(Continued)*

Crossing
Restriction to
Miss MOA

CROSSING RESTRICTION TO MISS MOA



“N342KG cleared to Texarkana Airport via direct Sidon Victor 278. Cross eight miles northeast Sidon VORTAC at or below seven thousand. Climb and maintain one zero thousand. B-G.”

32

NOTE: This animation is intended to illustrate the crossing restriction required to separate a GWO departure from the CBM MOA.

EXERCISE 2: DEMONSTRATING LATERAL SEPARATION BEST PRACTICES

Exercise 2



DEMONSTRATING LATERAL SEPARATION BEST PRACTICES EXERCISE



Purpose: to practice applying lateral separation rules

Directions: complete the flight strips based on the information provided by your instructor

33

Directions

In this exercise, you will practice stripmarking and phraseology to ensure lateral separation.

Your instructor will provide five separate sets of flight progress strips and lead the class through a discussion to determine the correct solution, phraseology, and stripmarking for each scenario.

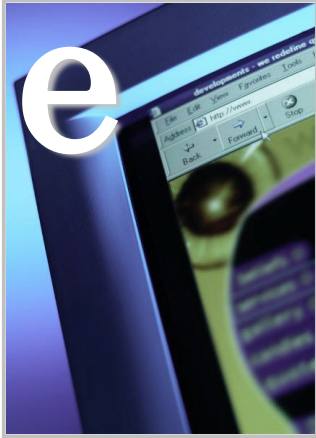
EXERCISE 2: DEMONSTRATING LATERAL SEPARATION BEST PRACTICES SCENARIOS 1-5

34-43

ACTIVITY: ENSURING SEPARATION

Activity

ENSURING SEPARATION ACTIVITY



Purpose: to identify and resolve conflicts by applying lateral separation rules

44

Description

In this activity, you will review flight scenarios with conflicts and then select the appropriate solutions to ensure lateral separation.

Directions

Access the IET eLearning menu. Select **Lesson 18 – Lateral Separation**. Click on the title to launch the **Ensuring Separation** activity.

Time Allotted

45 minutes

IN CONCLUSION

Lesson Review

LESSON REVIEW

The following topics were covered in this lesson:

- Lateral separation methods
- Protected airspace
- Minima on diverging radials



45

End-of-Lesson Test

END-OF-LESSON TEST

Lateral Separation



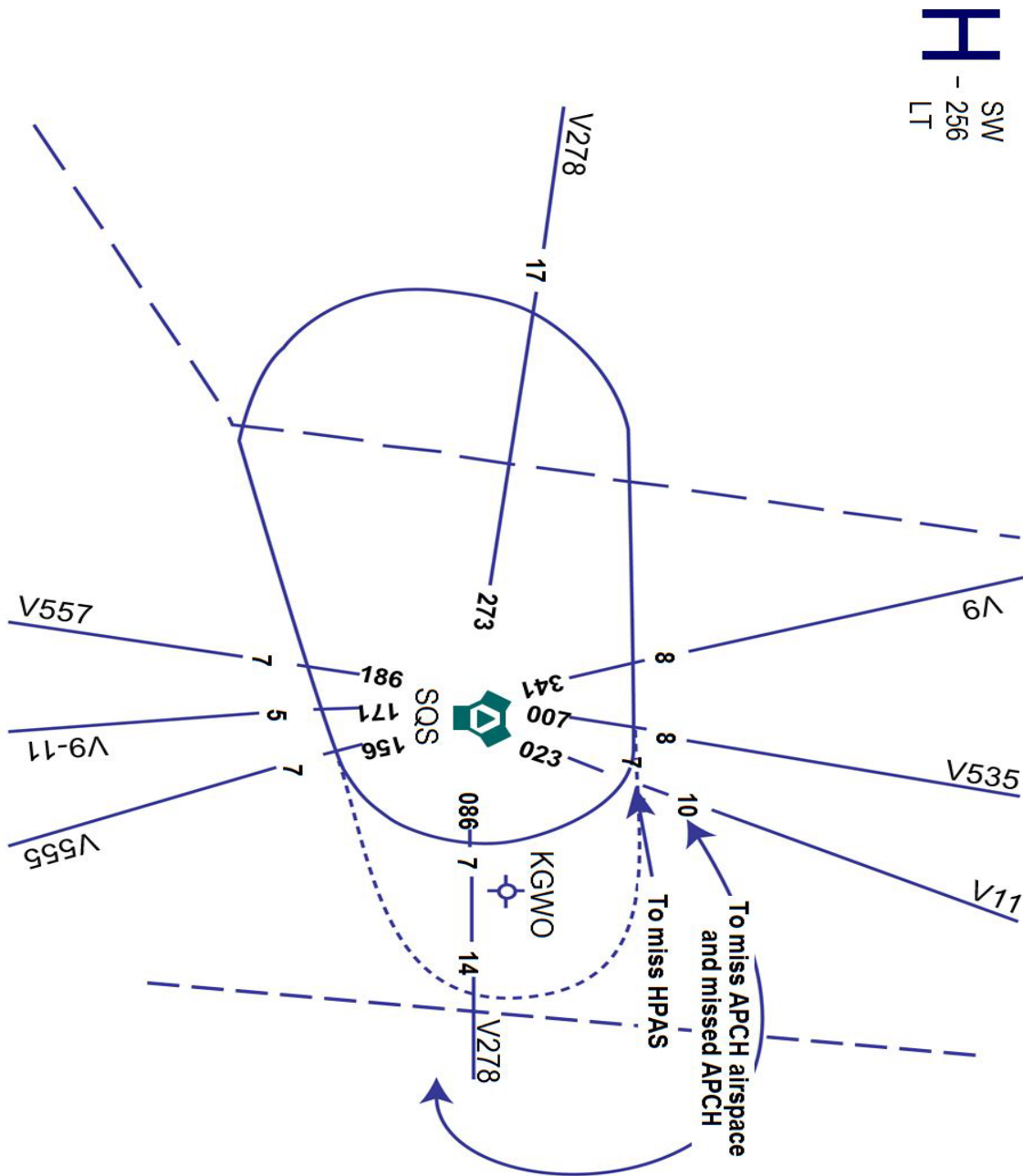
46

Part-Task Lab



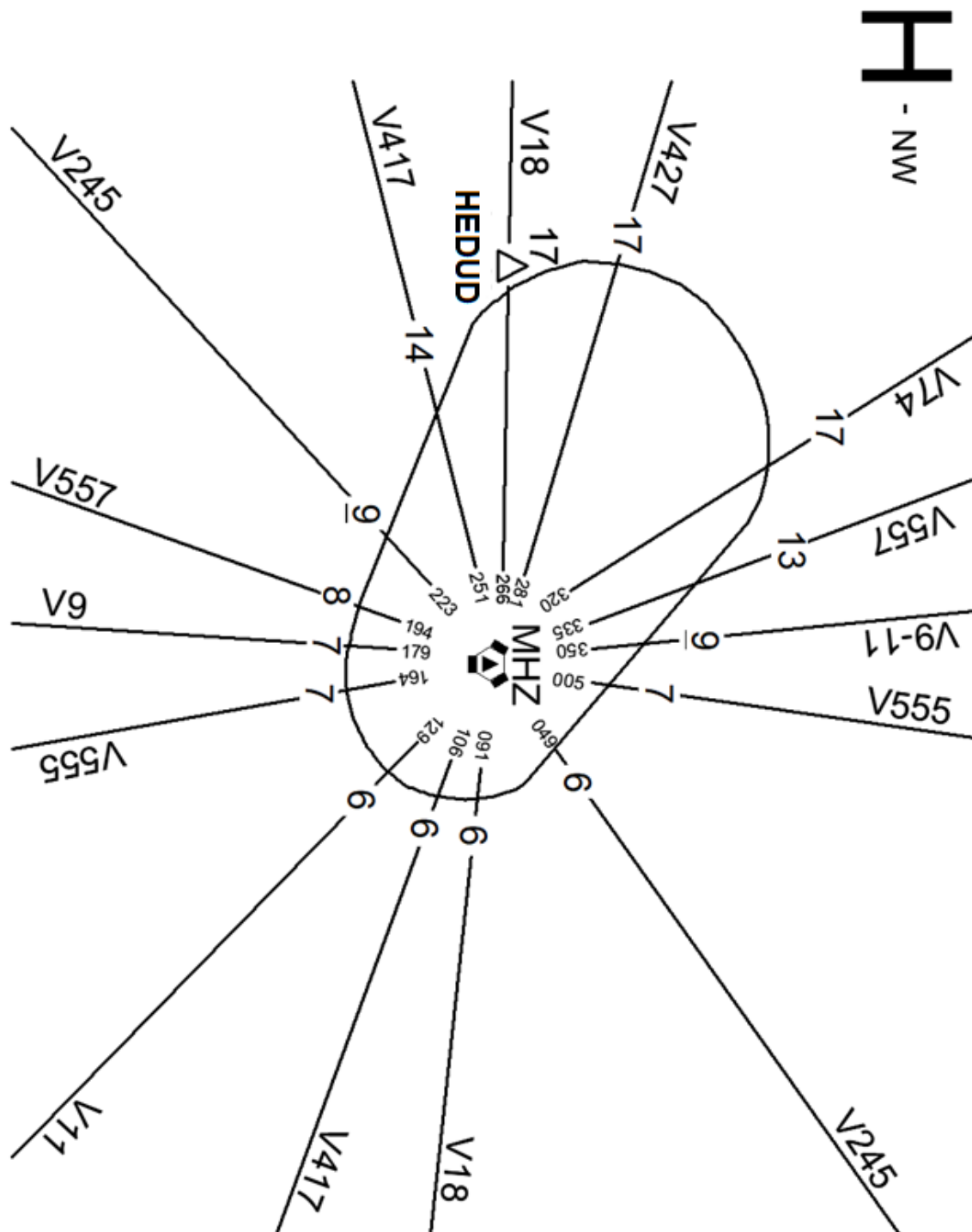
You will now complete the Lateral Separation part-task lab using LATPtask strips.

APPENDIX A: ZAE HOLDING PATTERNS

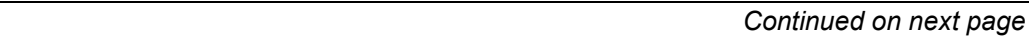


Continued on next page

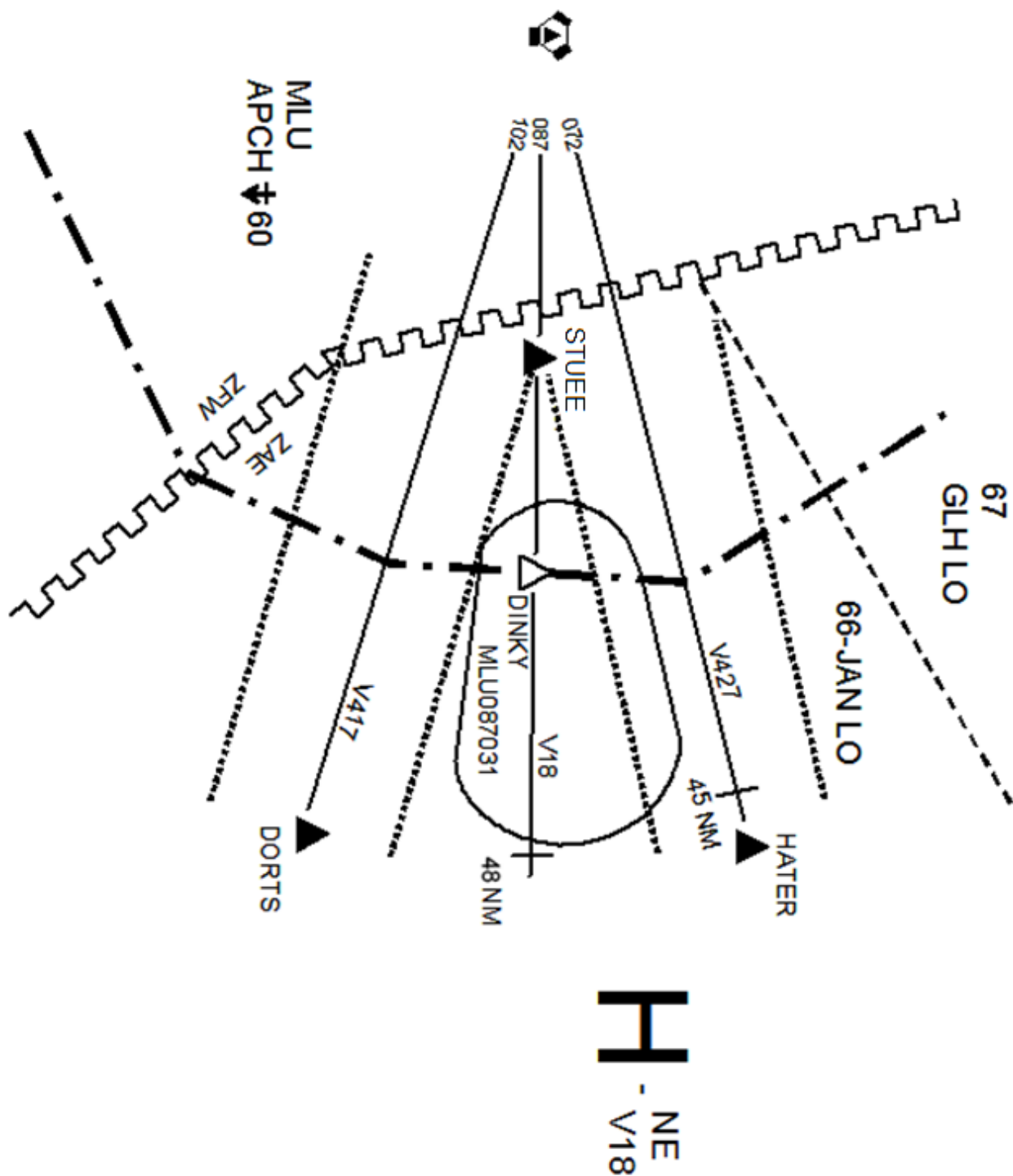
APPENDIX A: ZAE HOLDING PATTERNS *(Continued)*



Continued on next page



APPENDIX A: ZAE HOLDING PATTERNS *(Continued)*



Continued on next page